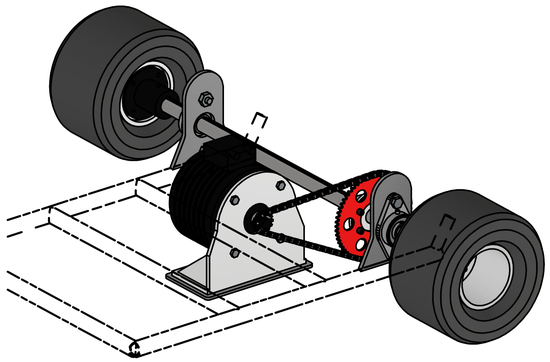
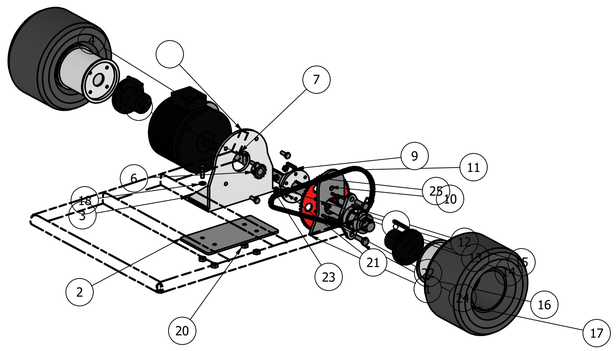
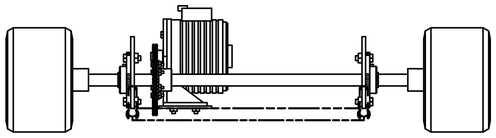
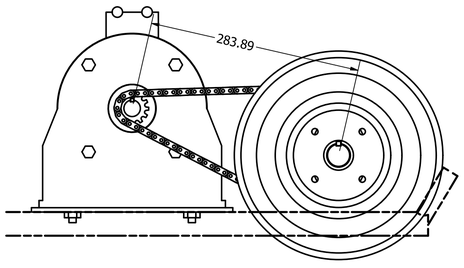




PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
14	2	GK-411	Rear wheel hub 1-1/4 bore x 80 long, 1/4 keyway, 4-bolt 90 mm flange, 6061-T6 (kart catalogue type)
15	2	GK-512	Square key 1/4 x 1/4 x 70, 1018 CD (wheel hubs)
16	2	GK-412	Kart rim 5 in dia x 5 in wide, cast aluminium (envelope)
17	2	GK-413	Kart slick 11 x 7.10-5 (rear), envelope
18	4	GK-528	Flat washer DIN 125 M10 (10.5 x 20 x 2), used under the 3/8 slot bolts
19	4	GK-521	Hex head cap screw 3/8-16 x 1-1/4, SAE J429 grade 8, zinc-yellow (motor-mount slot bolts)
20	4	GK-525	Nylon-insert locknut 3/8-16, grade 8
21	4	GK-522	Hex head cap screw 3/8-16 x 1, SAE grade 8 (motor face to mount plate)
22	6	GK-524	Socket head cap screw 5/16-18 x 1, ASTM A574 black oxide (sprocket to hub)
23	6	GK-527	Nylon-insert locknut 5/16-18, grade 8
24	4	GK-523	Hex head cap screw 1/2-13 x 1-1/2, SAE grade 8 (bearing flange to hanger)
25	4	GK-526	Nylon-insert locknut 1/2-13, grade 8

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	GK-101	Rear-axle bearing hanger, 3/8 in 4130 plate, R25 blends, saddled and TIG-welded to the main rail, 52 bore + 2 x 13.5
2	1	GK-102	Motor-mount base plate 270 x 142 x 6, A36, welded across the two 1-1/4 cross tubes, 4 x 10.5
3	1	GK-103	Motor mount bracket, 6061-T6 machined from solid: 3/8 in motor plate + 3/8 in base flange with 4 x 24 mm tensioning slots + 2 gussets
4	1	GK-201	Motenergy ME1114 brushless PMSM, 24-72 V, 0.12 Nm/A, 65 Nm peak, 5000 rpm, 7/8 shaft (catalogue envelope)
5	1	GK-513	Square key 3/16 x 3/16 x 19, 1018 CD (motor sprocket)
6	1	GK-401	Sprocket ANSI 35 13T, 7/8 bore, 3/16 key, B-hub (PD 39.80, OD 44.36), 1045, hardened teeth
7	1	GK-301	Rear live axle 1-1/4 in x 40 in, 4140 pre-hard 28-32 HRC, 1/4 x 1/8 full-length keyway, ends chamfered 1x45
8	1	GK-302	Axle sprocket hub 1-1/4 bore, 1/4 keyseat, flange 104 x 8, 6 x 8.4 on 3.25 in, 2 set screws, 6061-T6
9	1	GK-511	Square key 1/4 x 1/4 x 40, 1018 CD (sprocket hub)
10	1	GK-402	Kart sprocket ANSI 35 60T plate, 7075-T6 hard anodised, bore 64, 6 x 8.4 on 3.25 in (PD 182.00, OD 187.46)
11	1	GK-403	Roller chain ANSI 35 single strand, 98 links incl. connecting link (933.5 mm), min tensile 9.34 kN
12	2	GK-421	Flange housing UCFL206 2-bolt, cast iron, bolt centres 117 (catalogue dimensions)
13	2	GK-422	Insert ball bearing UC206-20, 1-1/4 bore, set-screw locking, C 19.5 kN, C0 11.3 kN





GOKARTDRIVETRAIN - ELECTRIC KART REAR DRIVETRAIN, 25 MPH (DESIGNED AND SIMULATED, NOT BUILT)
MOTOR MOTENERGY ME1114 PMSM 48 V, 0.12 NM/A, 350 A CONTROLLER = 42 NM; 16S LIFEPO4 40 AH
REDUCTION ANSI 35 CHAIN 13/60 T = 4.615:1, 98 LINKS, CENTRES 283.9 MM; 25 MPH = 3595 RPM MOTOR / 779 RPM AXLE
AXLE 1-1/4 IN 4140 PH LIVE, UC206-20 IN UCFL206 ON 3/8 IN 4130 HANGERS; TYRES 11X7.10-5, R ROLL 137 MM
SIMULATED 0-25 MPH 1.98 S (185 KG), CHAIN SF 4.4 ON MIN. ULTIMATE AT 42 NM
ASSEMBLY PARAMETERS: ChainLine (SPROCKET PLANE, Y) AND ChainAdjust (MOTOR SLOT, X)
EXPLODED VIEW = GoKartDrivetrain_Exploded.iam. THREADS COSMETIC. CHASSIS SECTION = REFERENCE (NOT IN PARTS LIST)

DRAWN Jay Patel	10/28/2026	
CHECKED		TITLE: GoKartDrivetrain - general assembly
QA		
MFG		
APPROVED		
		SIZE D · DWG NO GK-DWG-001 · REV
		SCALE 1 / 6 · SHEET 1 OF 1